# Introduction to Software Testing (ST) & Quality Assurance (QA)

## Course code: SEN304/SWE3202

## Why ST and QA Matter?

# Learning Objectives:



By the end of this lecture, you will be able to:

- Describe the software crisis and name two causes
- Distinguish Quality Assurance (QA) from Software Testing (ST).
- Relate one Nigerian case study to the consequences of poor testing.
- List the corresponding test phase for each development phase in the V-Model.

## The Software Crisis – What Was It?



**Timeframe**: Late 1960s – early 1970s

**Symptoms**: observed worldwide (NATO Conf., 1968):

"We were in a crisis: software was becoming impossible to manage."

- Projects running years behind schedule
- Massive cost/budget overruns
- Unreliable software that is full of bugs
- Difficult maintenance
- Adding more programmers made projects even later – "Brooks' Law" (Brooks, 1975)

# The IBM OS/360



Operating system for IBM's System/360 mainframe

**Scale:** Over 1,000 programmers, millions of lines of code (Brooks, 1975)

# What went wrong:

- No clear requirements – features kept changing
- Testing was an afterthought
- No quality processes. “it compiles” was the only check

## **Result:**

(Brooks, 1975; Charette, 2005):

- Delivered years late, far over budget
- Thousands of known bugs – IBM had to mail free patch tapes to every customer
- The crisis directly led to the birth of **software engineering** as a discipline.

# Could It Happen in Nigeria?

- Globally, the crisis never ended – it just changed shape.
- NITDA's finding (May 2025): More than half (56%) of IT projects in Nigerian Federal Public Institutions fail to meet their objectives.

# Common reasons

(NITDA IT Project Clearance Guidelines, 2025):

- No mandatory quality assurance phase
- Fragmented, uncoordinated development
- Lack of testing expertise before deployment

# Consequence:

- Financial losses
- Abandoned projects
- Public dissatisfaction

# Case Study 1:

The Abandoned Police Network (NPSCS)

**Project**: National Public Security Communication System (NPSCS) – a secure nationwide police communication network by ZTE/China Exim Bank, in 2008

**Investment:** Approximately $470 million (Premium Times, 2014; ICPC report, 2021)

# What failed:

- After launch, no funding for diesel or maintenance
- No quality control for long-term operation
- Facilities were abandoned; equipment looted

(ICPC, 2021; NASS Public Accounts Committee, 2022):

# Lesson learned:

A system is not successful just because it is deployed. Quality Assurance must include operations and maintenance planning.

13

# **Case Study 2** – Banking Sector Outage (2024)

**Incident**: A major Nigerian bank migrated to a new core banking application – “SEABaaS” (Sterling Bank, August 2024)

**Outcome**: over 3 million customers had delayed and failed transactions for two weeks (Nairametrics, Sept 2024; CBN oversight report)

**Root cause** (TechCabal, 2024; IT expert analysis):



- Inadequate pre-deployment testing
- No rollback plan in case of failure

**Impact**: Businesses crippled, public trust damaged

**Lesson learned:**

“A single untested banking application can shut down large parts of the economy.

# What Is Quality Assurance (QA)?

**QA** is about the process. It answers: “Are we building the product right?”

**Definition** (IEEE Std 610.12-1990):

“Planned and systematic activities to provide confidence that a product fulfills quality requirements.”

## QA activities happen before code is even written:



- Defining standards and guidelines
- Conducting design reviews
- Performing code inspections
- Auditing processes

**Goal of QA:** Prevent defects from being introduced.

# What Is Software Testing?

Testing is about the product. It answers: “Does the product work as required?”

**Definition** (IEEE Std 829-2008):

“The process of executing a system to find differences between existing and required behavior (i.e., bugs).”

Testing activities examine the running software:



- Executing test cases
- Comparing actual vs expected results
- Finding failures and reporting defects

**Goal of testing**: Detect defects that exist in the software

**QA** builds quality in; **testing** measures quality out. (Pressman, 2014)

# Quality Assurance (QA) vs Quality Control (QC) vs Testing



| Term | Focus | When it happens | Example |
|---|---|---|---|
| QA | Process | Throughout development | Reviewing requirements before coding |
| QC | Product | After a product is built | Checking if all tests passed |
| Testing | Execution | During/after coding | Running a test case and finding a bug |

**Source**: IEEE 610.12-1990; Pressman (2014)

**Bottom line:** You need all three. Skipping QA leads to chaotic testing (like OS/360).

# The Human Side and Economics of Testing

Psychology matters (Myers, 1979; Kaner et al., 1999):

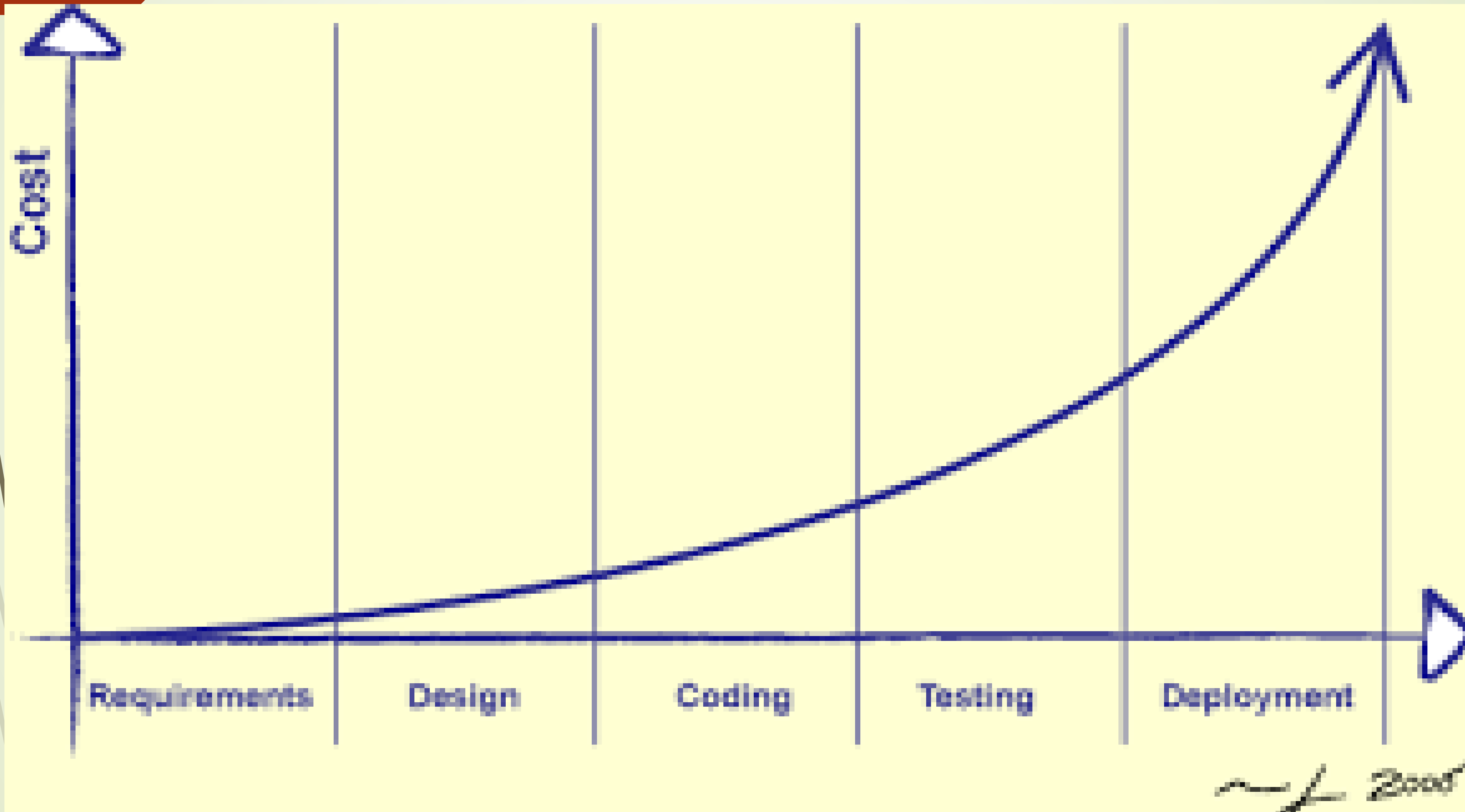
- The goal is to find bugs, not to prove the software works.
- A successful test is one that finds a failure.
- Testing should be constructive (i.e. aimed at improving the product, not blaming people).

## Economics of testing (Boehm, 1981; Pressman, 2014):

| Bug found at | Relative cost to fix |
|---|---|
| Requirements phase | 1× |
| Design phase | 3–6× |
| Coding phase | 10× |
| Testing phase | 20× |
| After release (production) | 50–100× |

**Bottom Line**: Finding a bug early saves enormous time and money.

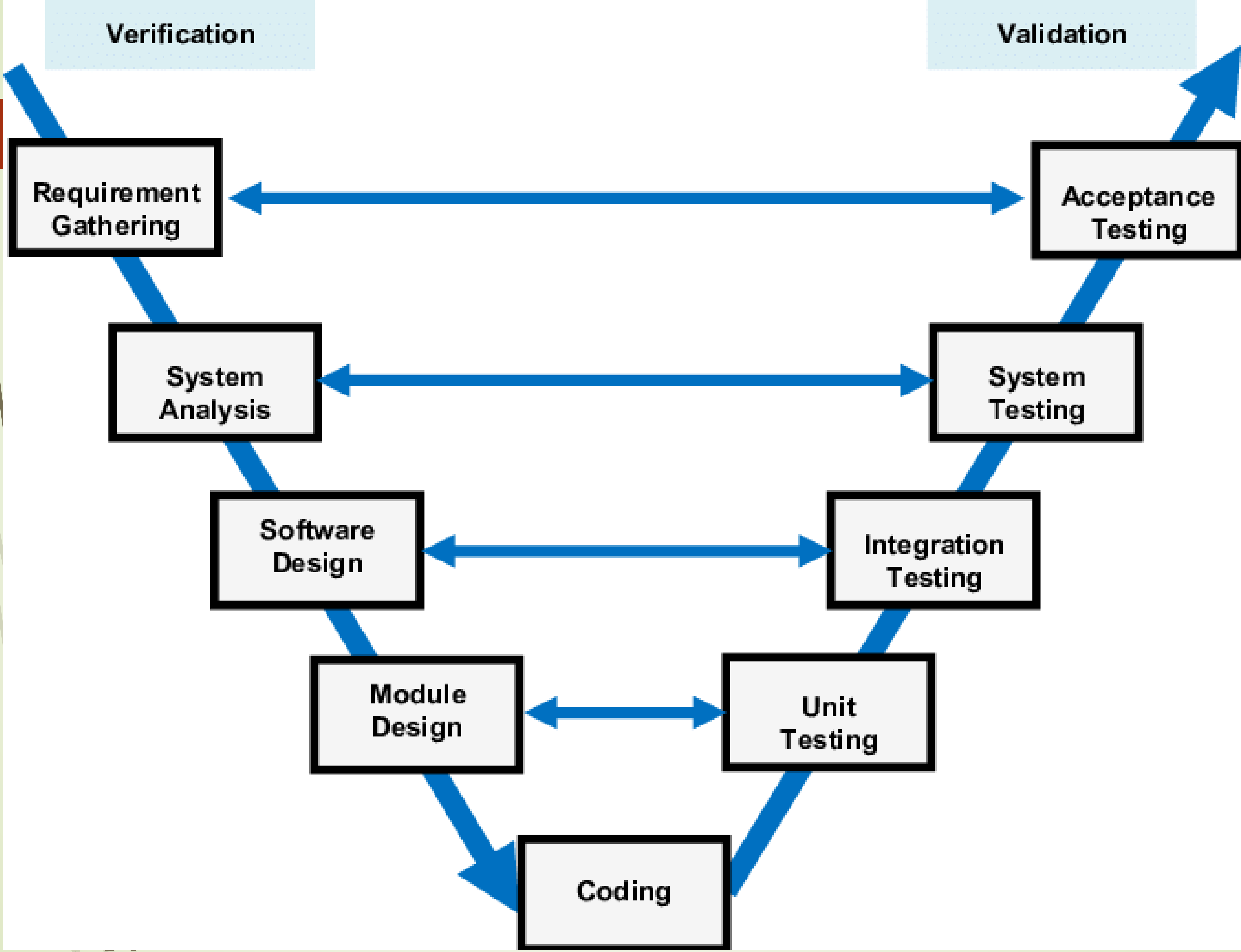
# BOEHM'S CURVE:




**Bottom Line**: Finding a bug early saves enormous time and money.

## The V-Model – Testing as a First-Class Citizen

The V-Model is a highly disciplined Software Development Life Cycle (SDLC) model that extends the Waterfall approach by pairing each development phase with a corresponding testing phase. It emphasizes early test planning to verify requirements, design, and code, aiming for high quality in projects with clear, fixed requirements.


Verification
Validation
Requirement Gathering
Acceptance Testing
System Analysis
System Testing
Software Design
Integration Testing
Module Design
Unit Testing
Coding

Traditional “code-then-test” is a recipe for crisis. The V-Model integrates testing from the start (ISO/IEC/IEEE 12207:2017).

**Principle**: For every development activity, a corresponding test activity is planned in parallel.

In the OS/360 project, testing was only planned after coding (Brooks, 1975), which was the core reason for it’s failure.

# Digital Security and Testing – The Nigerian Stake



**Cyber-related financial losses in Nigeria (2024):** Over N53.4 billion lost to cybercriminals (FITC Fraud and Forgery Report, Q1–Q3 2024)

How poorly tested software contributes (OWASP Top 10, 2021):

- Unpatched vulnerabilities
- Weak input validation (e.g., SQL injection)

Authentication flaws

# National experts (ANST) – Association of Nigeria Software Testers (2023, 2024):

"Without global-standard QA and testing, Nigeria will continue to suffer system glitches and security breaches."

(ANST President Demola Adesina, 2024)

Testing is not just a technical task – it is a national security and economic priority.

# The Shift – From “Afterthought” to “Essential”



**Old thinking** (1960s–1980s):

“We will write all the code first, then test it at the end.”

**New thinking** (industry standard – IEEE, ISO, ISTQB):

“Testing and QA activities begin on day one and continue throughout.”

# Our aim:

**This course will train you to be part of the solution.**

**You will learn how to design test cases, use automation, apply black-box and white-box techniques, and measure software quality.**

## Recap:



- The software crisis (IBM OS/360 – Brooks, 1975) proved unstructured development fails.
- The crisis is still with us – Nigerian projects fail at 56% (NITDA, 2025).
- QA prevents defects; testing detects defects.
- The V-Model ties testing to each development phase.
- Poor testing has real economic and security consequences in Nigeria (FITC 2024, NPSCS case).

"Based on the Nigerian banking outage case study (Sterling Bank, 2024), what three specific QA or testing activities would you have required before allowing the bank to go live? Explain why each would have helped."

# Thank you all.